

## **Simulator link:**

<https://hrfh-hackathon-2026.shinyapps.io/hackathon2026/>

## **Settings**

### **Simulation seed**

Controls the random variation in the simulated data.  
Using the same seed will reproduce the same dataset.

### **Simulation days**

Sets the number of days simulated after surgery.  
For example, 365 days represents one year of follow-up.

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## **Cohort sizes**

### **Fast recovery group**

Sets the number of participants in the fast recovery group.

### **Moderate recovery group**

Sets the number of participants in the moderate recovery group.

### **Slow recovery group**

Sets the number of participants in the slow recovery group.

These settings control the total number of simulated participants.

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## **Pre-surgery baseline**

### **Baseline variation**

Controls how different participants are before surgery.  
A low value means participants have similar pre-surgery activity levels.  
A high value means participants have more varied pre-surgery activity levels.

### **Fast group baseline change (%)**

Changes the pre-surgery activity level for the fast recovery group.

### **Moderate group baseline change (%)**

Changes the pre-surgery activity level for the moderate recovery group.

### **Slow group baseline change (%)**

Changes the pre-surgery activity level for the slow recovery group.

These settings allow recovery groups to start from different pre-surgery activity levels.

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## **BMI**

### **BMI value**

Sets the BMI value used for the simulated participants.

BMI affects the overall number of daily steps.

In this simulator, steps are highest in the healthy BMI range. Steps are lower when BMI is below or above this range.

BMI categories used in the app:

Underweight: below 18.5

Healthy: 18.5 to 24.9

Overweight: 25.0 to 29.9

Obese I: 30.0 to 34.9

Obese II: 35.0 to 39.9

Obese III: 40.0 or above

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## **Age**

### **Age**

Sets the age used for the simulated participants.

Age affects step count, recovery timing, and missingness.

In this simulator, older age lowers overall step count and delays recovery. Older age also leads to more complete data when missingness is applied.

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## **Long-term recovery level**

### **Fast group long-term level**

Controls the final activity level reached by the fast recovery group.

### **Moderate group long-term level**

Controls the final activity level reached by the moderate recovery group.

### **Slow group long-term level**

Controls the final activity level reached by the slow recovery group.

A value of 1 means returning to the pre-surgery activity level.

A value above 1 means activity becomes higher than the pre-surgery level.

A value below 1 means activity remains lower than the pre-surgery level.

### **Between-person variation in long-term level**

Controls how much participants differ in their final recovery level.

A low value means participants in the same group have similar long-term recovery levels.

A high value means participants in the same group have more varied long-term recovery levels.

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## **Recovery timing**

### **Fast group recovery speed**

Controls how quickly the fast recovery group improves.

Higher values mean faster recovery.

### **Moderate group recovery speed**

Controls how quickly the moderate recovery group improves.

Higher values mean faster recovery.

### **Slow group recovery speed**

Controls how quickly the slow recovery group improves.

Higher values mean faster recovery.

### **Fast group recovery midpoint (days)**

Sets the approximate day when the fast group reaches the middle of its recovery curve.

### **Moderate group recovery midpoint (days)**

Sets the approximate day when the moderate group reaches the middle of its recovery curve.

### **Slow group recovery midpoint (days)**

Sets the approximate day when the slow group reaches the middle of its recovery curve.

A smaller midpoint means recovery happens earlier.

A larger midpoint means recovery happens later.

### **Between-patient variation in recovery timing**

Controls how much recovery timing differs between participants.

A low value means participants recover at similar times.

A high value means some participants recover much earlier or later than others.

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## **Missing data**

### **Missing data level**

Sets the target percentage of missing step-count values.

For example, 50% means that about half of the step-count values are removed.

### **Missingness data seed**

Controls the random missingness pattern.

Using the same seed will reproduce the same missing data pattern.

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## **Random gaps: MCAR**

MCAR means Missing Completely At Random.

In this setting, missing data are introduced randomly. The probability of missingness does not depend on time, participant characteristics, or activity level.

### **Gap length distribution**

Controls whether random missing data appear as short or long gaps.

A low value creates more short gaps.

A high value creates more long gaps.

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## More missing early in recovery: MAR

MAR means Missing At Random.

In this setting, missingness depends on observed information, such as time after surgery, BMI, and age.

### Early recovery effect

Controls whether missing data are more common early after surgery.

A low value spreads missing data more evenly across time.  
A high value concentrates missing data early in recovery.

### BMI and age effect

Controls how strongly BMI and age influence missingness.

A low value gives BMI and age little effect.  
A high value gives BMI and age a stronger effect.

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## Low-activity days more likely missing: MNAR

MNAR means Missing Not At Random.

In this setting, low-activity days are more likely to be missing.

### Low-activity sensitivity

Controls how strongly low step-count values increase the chance of missingness.

A low value gives low-activity days little extra missingness risk.  
A high value makes low-activity days much more likely to be missing.

### Time window

Sets the time window used to identify lower-activity days.

Within each time window, days below the window average are more likely to be missing.

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## Example participant selection

These settings appear after running the simulation.

### **Fast group**

Selects up to three example participants from the fast recovery group.

### **Moderate group**

Selects up to three example participants from the moderate recovery group.

### **Slow group**

Selects up to three example participants from the slow recovery group.

These selections allow users to inspect individual recovery patterns.

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## **Missing data comparison selection**

These settings appear after running the simulation.

### **Fast group**

Selects one fast-recovery participant for missing data comparison.

### **Moderate group**

Selects one moderate-recovery participant for missing data comparison.

### **Slow group**

Selects one slow-recovery participant for missing data comparison.

These selections allow users to compare how MCAR, MAR, and MNAR missingness affect the same participants.

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## **Dataset preview and download**

### **Dataset to preview or download**

Selects which dataset to view or download.

Available options are:

1. Complete data
2. MCAR missing data
3. MAR missing data

4. MNAR missing data
5. All datasets as a ZIP file